Exercise 1

1. Given the data in Table 1 for an NMOS transistor with $k'=20\mu A/V^2$, calculate V_{T0}, λ, and W/L.

Table 1: Measured NMOS transistor data

	V_{GS}(V)	V_{DS}(V)	V_{SB}(V)	I_D(μA)
1	3	5	0	1210
2	5	5	0	4410
3	5	10	0	5292

2. Consider the circuit configuration in Figure 1.

Write down the equation (and only those) which you need to determine the voltage at node x. Assume that $\lambda_P=0$.

Determine the required width of the transistor (for L_{eff}=.25μm) such that x equals 1.5V.

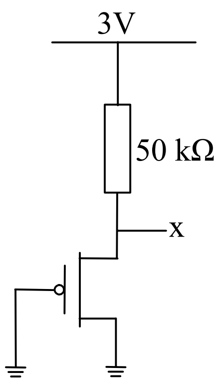


Figure 1: MOS circuit.

3. Following transistor values for a typical process are given: $V_T = 0.53$ V, $\gamma = 0.35$, $V_{DSAT} = 0.75$ V, $k = 145*10^{-6}$ A/V^2, $\lambda = 0.07$ 1/V

The drain current I_D of a NMOS transistor is defined as

$$I_D = k\left((V_{GS} - V_T)V_{MIN} - \frac{V_{MIN}^2}{2} \right)(1 + \lambda V_{DS})$$

with

$$V_{MIN} = \min(V_{GS} - V_T, V_{DS}, V_{DSAT})$$

3a. Draw the I_D *versus* V_{DS} characteristics for following values

V_{DS}= 0; 0,5; 0,75; 1; 1.5; 2; 2.5 (all values in V)
V_{GS}= 1.0; 1.75; 2.5 (all values in V)

3b. What type of device is it a short- or a long-channel device? Give a **short** motivation.

3c. In the technology development towards lower feature sizes follows lower supply voltages (V_{DD}). This also means that the voltage between V_{DD} and the threshold voltage (V_T) is decreasing, which leads to lower gain and longer propagation delay. One method to compensate for that is to use a technology that supports lower threshold voltages. What is the major problem with decreasing the threshold voltage?

3d. As a circuit designer, you can decrease the threshold voltage. How?

4. The following MOSFET transistor parameters for a typical technology are given

$V_{T0} = -0.5$ V
$V_{DSAT} = -1.0$ V
$k = -45*10^{-6}$ A/V^2
$\lambda = -0.09$ 1/V
$\gamma = -0.4$
$2\Phi_F = 0.6$V

The drain current I_D of a MOS transistor is defined as

$$I_D = k\left((V_{GS} - V_T)V_{MIN} - \frac{V_{MIN}^2}{2} \right)(1 + \lambda V_{DS})$$

with

$$V_{MIN} = \min(V_{GS} - V_T, V_{DS}, V_{DSAT})$$

4a. Two types of MOSFET devices are used in CMOS technologies. What type of transistor device is described by the parameters above? Give a **short** motivation.

4b. If the transistor is placed in a well, what type of well should it be? Give a **short** motivation.

4c. Draw the I_D/V_{DS} characteristics for the following values.

V_{DS} = 0, -0,5, -0,75, -1, -1.5, -2, and -2.5 Volts

V_{GS} = -1.0, -1.75, and -2.5 Volts

4d. What type of device is it a short- or a long-channel device? Give a **short** motivation.

4e. The well where the device is placed, is connected to 3 V, while the supply voltage V_{DD} = 2.5 V. Determine the threshold voltage V_T.

5. The drain current plot for an *n*- and a *p*-channel transistor is shown in Figure 2a and 2b, respectively.

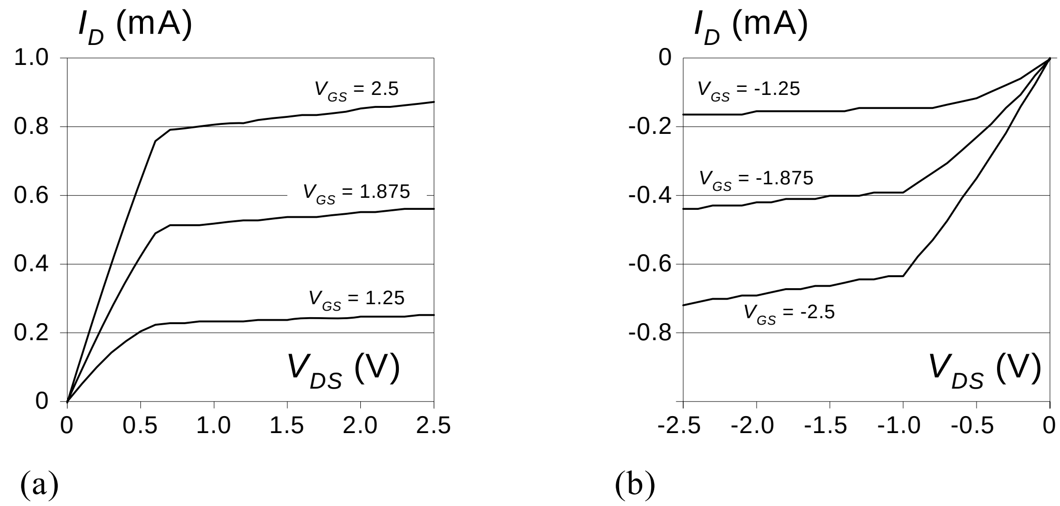


Figure 2: The drain current vs. the drain-source voltage

5a. The drain currents for the corresponding minimal transistors are 0.22 mA and -0.09 mA, respectively (for $V_{DS} = V_{GS} = 2.5$ V). Determine the transistor widths (W_n and W_p) for the transistors in Figure 2 if the minimal width is 0.375 μm and the lengths are remaining unchanged.

5b. Determine the average on-resistances, $R_{eq\text{-}n}$ and $R_{eq\text{-}p}$, for the two transistors in Figure 2.

5c. Use the two transistors in an inverter and determine the propagation delays t_{pHL} and t_{pLH} when the inverter is loaded with 2 pF.

6

Figure 3 shows a clock distribution on a chip. The clock wires are fed from a large clock buffer. Four small buffers are terminating the clock distribution. The length of each piece of wire is shown in the figure. The widths of the wires are 0.4 µm and the sheet resistance is 0.1 Ω/□ (ohms per square). The area and fringe capacitance have the values 0.03 fF/µm^2 and 0.05 fF/µm respectively. The inter-wire capacitance is 0.1 fF/µm, between two parallel wires. The small buffers at the end of the wires have a load C_L of 1 pF each. The input buffer load is neglected. Assume a supply voltage at 2.5 volt.

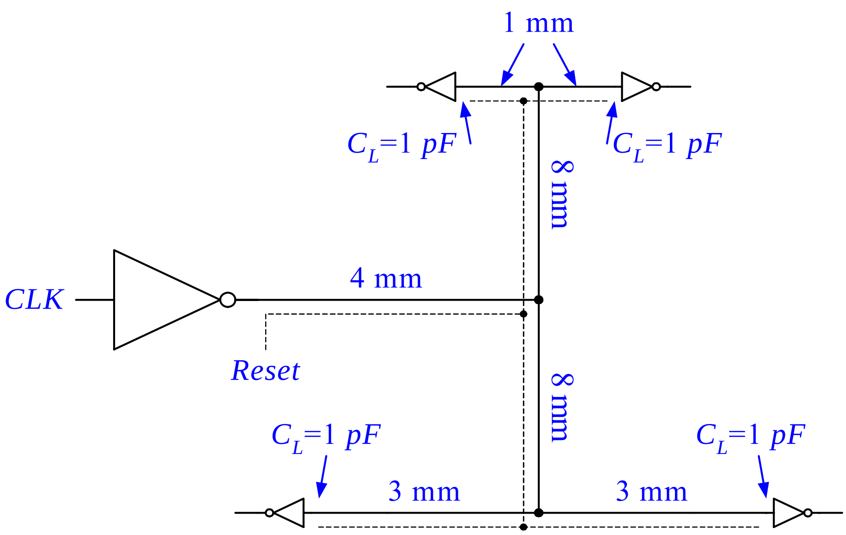


Figure 3 A two level clock distribution net.

6a. Determine the relative clock delay at the end of the distribution (the delay difference between the clock arrival at the end of the upper and the lower paths), by applying the lumped RC model for the whole clock net. Use the time to reach the 50% point ($V_{DD}/2$) in the delay calculation. Neglect the interwire capacitance.

6b. A reset signal is routed in parallel with the clock net, in the same metal layer (the dashed wire). Determine the voltage drop on the reset wire, which is considered to be in a high impedance mode, when *Reset* is high and the clock wire does an ideal transition from high to low. Neglect the wire resistance.

6c. Determine the relative clock delay in the clock distribution, by applying the Elmore delay on each piece of wire. Assume that the lumped RC model can be used for each piece of wire. Neglect the interwire capacitance

7. An n-channel MOS transistor with the dimensions W/L = 0.6/0.3 have the I-V characteristics shown in Figure 4.

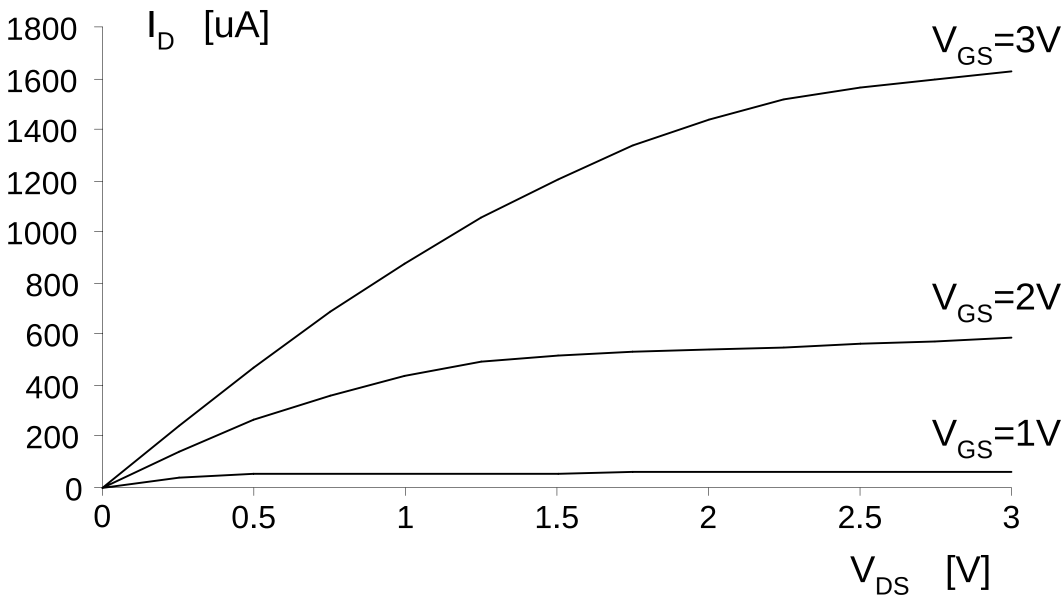


Figure 4: I_D as a function of V_{DS}

7a. Is the transistor velocity saturated? Give a short motivation!

7b.To determine λ graphically, the derivative of I_D regarding V_{DS} is calculated

$$\frac{\partial I_D}{\partial V_{DS}} = \lambda \frac{k'_n}{2} \frac{W}{L} (V_{GS} - V_T)^2 \approx \lambda I_D$$

$$\lambda \approx \frac{1}{I_D} \frac{\Delta I_D}{\Delta V_{DS}}$$

Determine an approximation of λ graphically.

7c. Determine the threshold voltage V_T using an appropriate graph.

7d. Determine the gain factor k_n'.

8. Figure 5 shows a layout of a dynamic memory cell using n-channel transistors. This cell is called the three-transistor cell or the three-transistor dynamic memory cell.

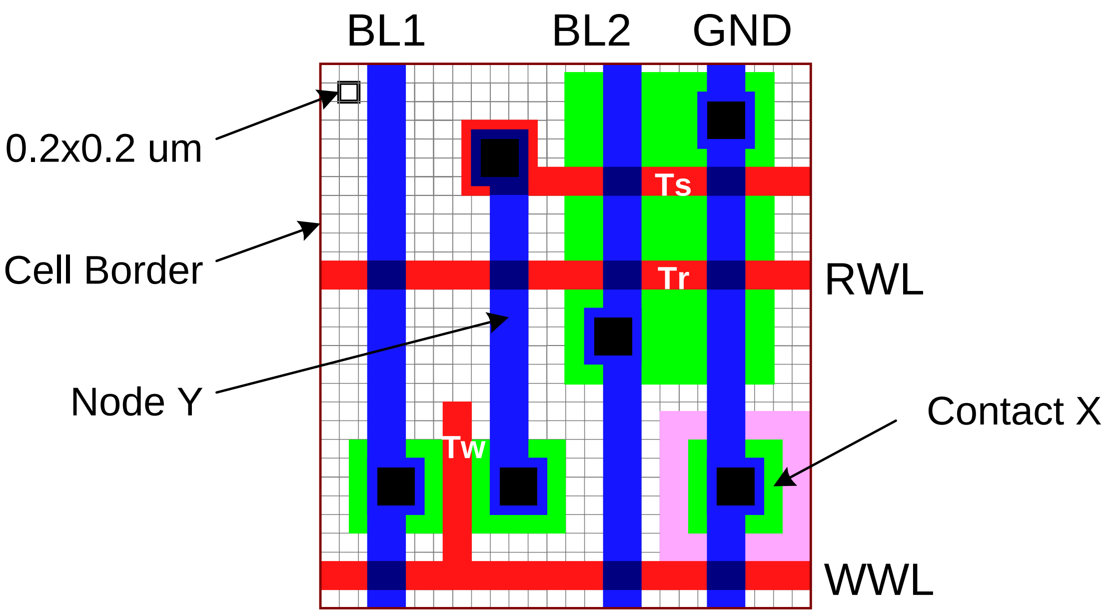


Figure 5: A dynamic memory cell

Calculate the capacitance in node Y if it consists of the gate capacitance in Ts and the drain capacitance in Tw. C_{ox} = 5 fF/um^2, C_{j0} = 1.5 fF/um^2, and C_{jsw0} = 0.25 fF/m

Solutions 1

1.

$k' = 20\mu A/V^2$

If $V_{DS} > V_{GS} - V_t$, then eq. 3.29 can be used (in saturation).

To find V_{t0}, find two measures in saturation, having the same V_{DS} and $V_{BS}=0$.

Measure no 1: $V_{DS1}=5V$ $V_{GS1}=3V$. If $V_t > -2V$, in saturation $\Rightarrow$ eq. 3.29 can be used.

Measure no 2: $V_{DS2}=5V$ $V_{GS2}=5V$. If $V_t > 0V$, in saturation $\Rightarrow$ eq. 3.29 can be used.

Assume $V_t > 0V$ (Mostly true).

Eq. 3.29$\Rightarrow$ $$\frac{(V_{GS1} - V_{t0})^2}{(V_{GS2} - V_{t0})^2} = \frac{I_{D1}}{I_{D2}} \Rightarrow \ldots\ldots \Rightarrow V_{t0} = \frac{V_{GS2}\sqrt{\frac{I_{D1}}{I_{D2}}} - V_{GS1}}{\left(\sqrt{\frac{I_{D1}}{I_{D2}}} - 1\right)} = 0.8V$$

To find λ, find two measures in saturation, having the same V_{GS} and $V_{BS}=0$.

Measure no 2: $V_{DS2}=5V$ $V_{GS2}=5V$. If $V_t > 0V$, in saturation $\Rightarrow$ eq. 3.29 can be used.

Measure no 3: $V_{DS3}=10V$ $V_{GS3}=5V$. If $V_t > -5V$, in saturation $\Rightarrow$ eq. 3.29 can be used.

Eq. 3.29$\Rightarrow$ $$\frac{(1+\lambda V_{DS2})}{(1+\lambda V_{DS3})} = \frac{I_{D2}}{I_{D3}} \Rightarrow \ldots\ldots \Rightarrow \lambda = \frac{\left(\frac{I_{D2}}{I_{D3}} - 1\right)}{\left(V_{DS2} - \frac{I_{D2}}{I_{D3}} V_{DS3}\right)} = 0.05V^{-1}$$

Use any measure in saturation with $V_{BS}=0$ to find W/L.

For no. 2.

Eq. 3.29$\Rightarrow$ $$\frac{W}{L} = \frac{2I_{D2}}{k_n'(V_{GS2} - V_t)^2(1+\lambda V_{DS2})} = 20$$

2. Use Process parameters of Table 3.2 (page 103)

$$\begin{cases} I_D = \dfrac{k'_p}{2}\dfrac{W}{L}\left(V_{GS} - V_t\right)^2 = \dfrac{k'_p}{2}\dfrac{W}{L}\left(-V_X + 0.4\ \right)^2 = -72.6W \\ I_D = \dfrac{V_X - 3}{50 \cdot 10^3} = -30 \cdot 10^{-6} \end{cases}$$

W = 0.41µm.

3.

The transistor values are V_T=0.53, γ=0.35, V_{DSAT}=0.75 , k=145*10^{-6}, λ=0.07, and the drain current is

$$I_D = k\left(\left(V_{GS} - V_T\right)V_{MIN} - \frac{V_{MIN}^2}{2}\right)\left(1 + \lambda V_{DS}\right)$$

with

$$V_{MIN} = \min\left(V_{GS} - V_T, V_{DS}, V_{DSAT}\right)$$

3a. The drain current versus the drain-source voltage is plotted in Figure 1 for

V_{DS}= 0; 0,5; 0,75; 1; 1.5; 2; 2.5
V_{GS}= 1.0; 1.75; 2.5

3b. It is a short channel device. The drain current saturates at 0.75 V

3c. The subthreshold current is increasing exponentially leading to higher static power consumption.

3d. By increasing the base voltage, .i.e. decrease V_{SB} , the threshold voltage will decrease according to

$$V_T = V_{T0} + \gamma(\sqrt{\left|-2\phi_F + V_{SB}\right|} - \sqrt{\left|-2\phi_F\right|})$$

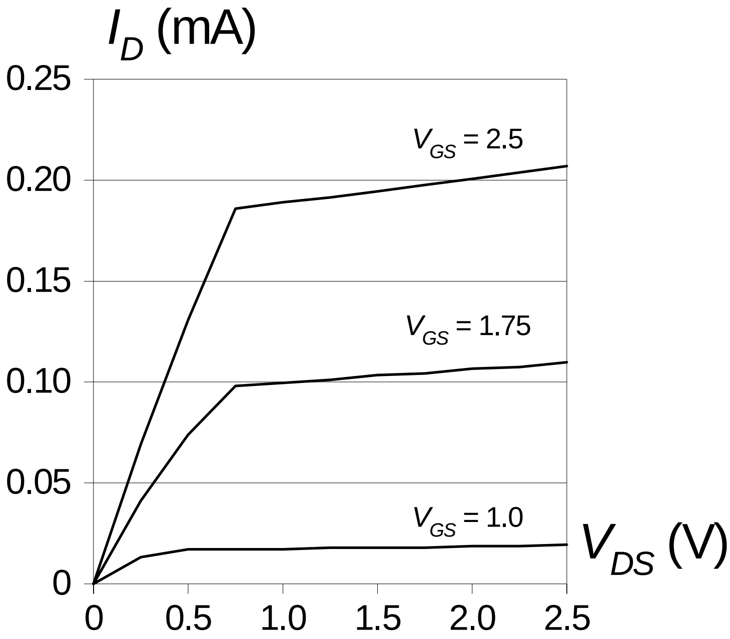


Figure 1: The drain current vs. the drain source voltage

4.

The transistor values are

V_{T0} = -0.5 V
V_{DSAT} = -1.0 V
k = -45*10^{-6} A/V^2
λ = -0.09 1/V
γ = -0.4
2Φ_F = 0.6V

and the drain current is

$$I_D = k\left((V_{GS} - V_T)V_{MIN} - \frac{V_{MIN}^2}{2} \right)(1 + \lambda V_{DS})$$

with

$$V_{MIN} = \min(V_{GS} - V_T, V_{DS}, V_{DSAT})$$

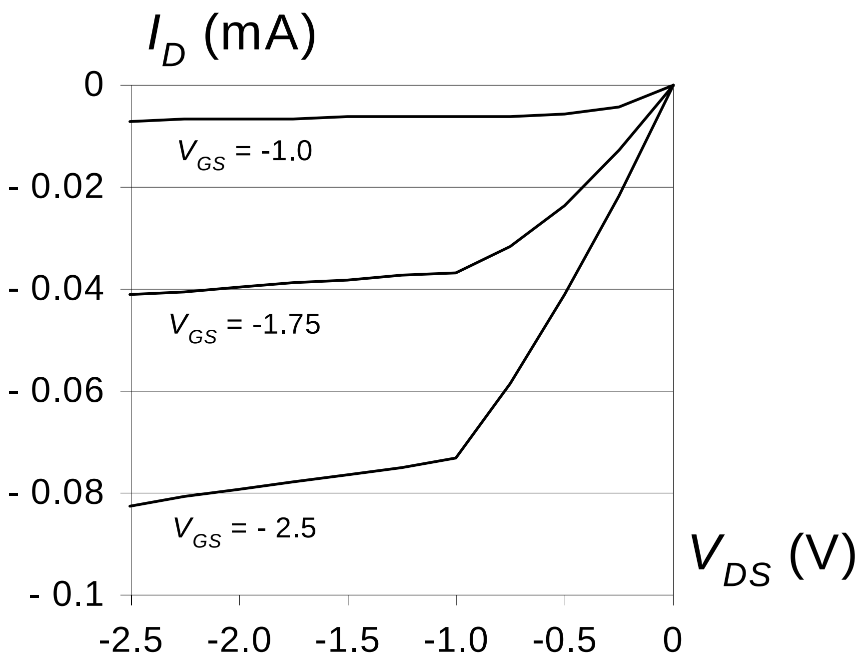


Figure 2 The drain current vs. the drain source voltage

4a Since most of the parameters are negative the transistor must be a *p*-channel device.

Answer: It is a PMOS transistor

4b. . Since it is a PMOS transistor, the well must be *n*-doped.

Answer: *n*-well

4c. The drain current versus the drain-source voltage is plotted in Figure 2.

4d. It is a short channel device since the drain current saturates at – 1.0 V

4e. The new threshold voltage is determined by

$$V_T = V_{T0} + \gamma(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|-2\phi_F|})$$

where

$$V_{SB} = V_{DD} - V_{Well} = 2.5 - 3.0 = -0.5 \text{ V}$$

which yields

$$\begin{aligned} V_T &= V_{T0} + \gamma(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|-2\phi_F|}) = \\ &= -0.5 - 0.4(\sqrt{|-0.6-0.5|} - \sqrt{|-0.6|}) = \\ &= -0.5 - 0.4(0.2742) = -0.6097 \end{aligned}$$

5.
The drain currents for the given transistors are about 0.88 and -0.72 mA respectively, at V_{DS}=V_{GS}=2.5 V. The currents are about 0.82 and -0.64 mA respectively, at V_{DS}=1.25 and V_{GS}=2.5 V

5a. The NMOS is thus four times wider and the PMOS is eight times wider than the minimum width, i.e., W_n = 1.5 and W_p = 3.0 µm.

5b. With Equation 3.41 (page 104) and the values of the drain currents at V_{DD} and $V_{DD}/2$ we get

$$\mathrm{R}_{eq-n} = \frac{\dfrac{V_{DD}}{I_{Dn}\big|_{V_{DD}}} + \dfrac{\frac{V_{DD}}{2}}{I_{Dn}\big|_{V_{DD}/2}}}{2} = \frac{\dfrac{2.5}{0.88\times10^{-3}} + \dfrac{1.25}{0.82\times10^{-3}}}{2} = 2.2\ \text{k}\Omega$$

$$\mathrm{R}_{eq-p} = \frac{\dfrac{V_{DD}}{I_{Dp}\big|_{V_{DD}}} + \dfrac{\frac{V_{DD}}{2}}{I_{Dp}\big|_{V_{DD}/2}}}{2} = \frac{\dfrac{-2.5}{-0.72\times10^{-3}} + \dfrac{-1.25}{-0.64\times10^{-3}}}{2} = 2.7\ \text{k}\Omega$$

5c. The propagation delays are (page 206)

$$t_{pHL} = 0.69R_{eq-n}C_L = 0.69\times2200\times2\times10^{-12} = 3.0\ \text{ns}$$
$$t_{pLH} = 0.69R_{eq-p}C_L = 0.69\times2700\times2\times10^{-12} = 3.7\ \text{ns}$$

6
A clock distribution net is shown. The general idea is to find equally long paths to minimize the relative delay (the relative clock skew). The wire resistance, R, capacitance to ground C, and interwire capacitance C_I per length unit are

$$R = \frac{L}{W}R_{\square} = \frac{1000}{0.4}0.1 = 250\ \Omega/\text{mm}$$
$$C = C_{Area} + 2\times C_{Fringe} = 1000\times0.40\times0.03 + 2\times1000\times0.05 = 112\ \text{fF}/\mu\text{m}$$
$$C_I = 1000\times0.1 = 100\ \text{fF}/\mu\text{m}$$

6a. To determine the delay time we need to know the resistance from the input buffer to each end point of the wires and the total capacitance for the whole clock net. The resistances and the capacitance are thus determined by (the interwire capacitance is neglected here)

$$L_{Short} = 4 + 8 + 1 = 13 \text{ mm}$$
$$L_{Long} = 4 + 8 + 3 = 15 \text{ mm}$$
$$L_{Total} = 4 + 2\times 8 + 2\times 3 + 2\times 1 = 28 \text{ mm}$$
$$R_{Short} = L_{Short} \times R = 13\times 250 = 3250\,\Omega$$
$$R_{Long} = L_{Long} \times R = 15\times 250 = 3750\,\Omega$$
$$C_{Total} = L_{Total} \times C + 4\times C_{Buffer} = 28\times 112 + 4\times 1000 = 7136 \text{ fF}$$

The absolute and relative delays are given by

$$t_{Long} = 0.69\times R_{Long} \times C_{Total} = 0.69\times 3750\times 0.007136 = 18.4644 \text{ ns}$$
$$t_{Short} = 0.69\times R_{Short} \times C_{Total} = 0.69\times 3250\times 0.007136 = 16.025 \text{ ns}$$
$$t_{Skew} = 18.4644 - 16.025 = 2.46 \text{ ns}$$

Answer: The relative clock delay is 2.5 ns

A simplified method would be to use the length difference only

$$L_{Extra} = L_{Long} - L_{Short} = 2 \text{ mm}$$
$$t_{Skew} = t_{Long} - t_{Short} = 0.69\times((L_{Short} + L_{extra}) - L_{Short})\times R\times C_{Total} =$$
$$= 0.69\times L_{extra} \times R\times C_{Total} = 0.69\times 2\times 250\times 0.007136 = 2.46 \text{ ns}$$

6b. The inter-wire capacitance is calculated with

$$C_{Interwire} = L_{Total} \times C_I = 28\times 100 = 2800\ fF$$

The voltage drop is determined by capacitive voltage division (p 446)

$$V_{Drop} = \frac{C_{Interwire}}{C_{Interwire} + C_{Total}} V_{DD} = \frac{2800}{2800 + 7136}\times 2.5 = 0.7045\,V$$

Answer: 0.70 V

6c. The Figure shows the an RC network equivalent to the H-tree

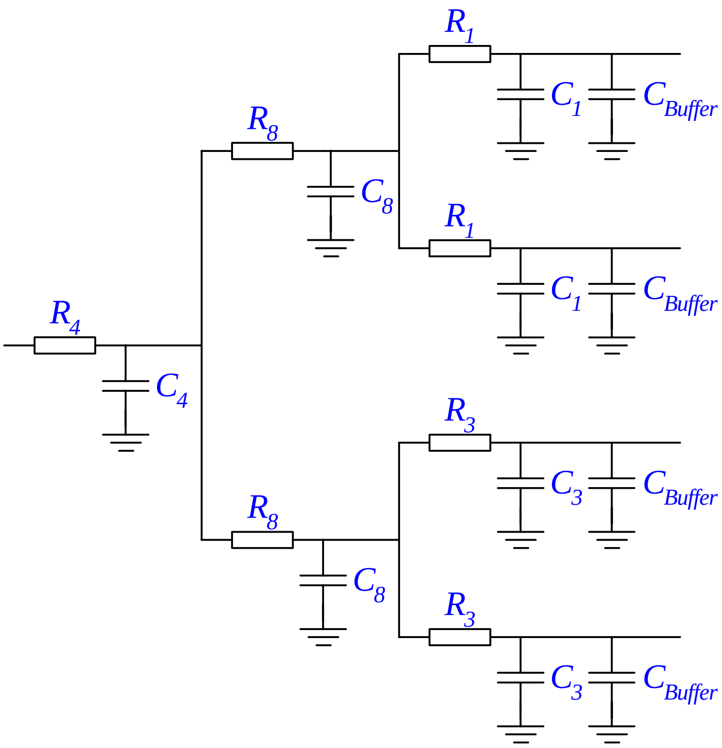


Figure Tree structured RC network

The resistances are determined by

$$R_4 = 4R = 1000\ \Omega$$
$$R_8 = 8R = 2000\ \Omega$$
$$R_1 = R = 250\ \Omega$$
$$R_3 = 3R = 750\ \Omega$$

The area and fringing capacitances are

$$C_4 = 4C = 448\ fF$$
$$C_8 = 8C = 896\ fF$$
$$C_1 = C = 112\ fF$$
$$C_3 = 3C = 336\ fF$$

The absolute and relative delays are given by (p 154):

$$t_{short} = 0.69\times(R_4\times C_4 + (R_4+R_8)\times C_8 + (R_4+R_8+R_1)\times(C_1+C_{buffer}) +$$
$$+(R_4+R_8)\times(C_1+C_{buffer}) + R_4\times C_8 + R_4\times(C_3+C_{buffer}) + R_4\times(C_3+C_{buffer})) =$$
$$= 0.69\times R\times(193C+33C_{buffer}) = 9.42ns$$

$$t_{long} = 0.69\times(R_4\times C_4 + (R_4+R_8)\times C_8 + (R_4+R_8+R_3)\times(C_3+C_{buffer}) +$$
$$+(R_4+R_8)\times(C_3+C_{buffer}) + R_4\times C_8 + R_4\times(C_1+C_{buffer}) + R_4\times(C_1+C_{buffer})) =$$
$$= 0.69\times R\times(233C+35C_{buffer}) = 10.54ns$$

Answer: 1.12 ns

7.

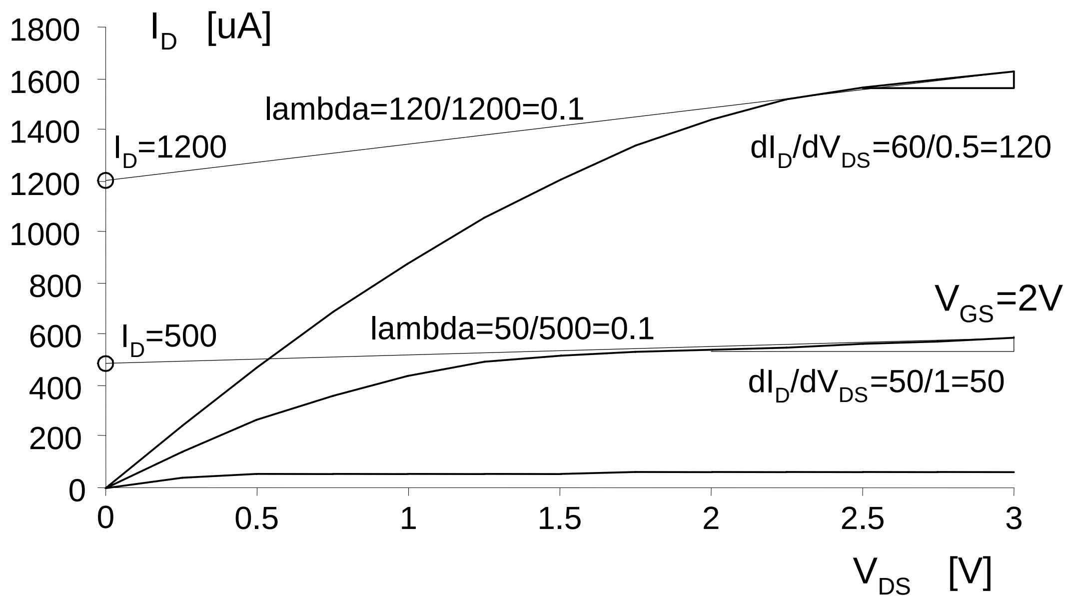
7a. No, since there is a quadratic I_D-V_{GS} relationship and not a linear.

7b. λ is determined by $\Delta I_D/\Delta V_{GS}$ in the saturated region together with I_D when V_{DS}=0.

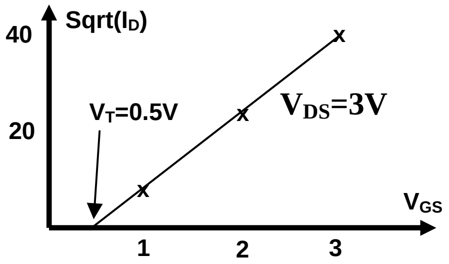
$$\frac{\delta I_D}{\delta V_{DS}} = \lambda\times\underbrace{\frac{k'_n}{2}\frac{W}{L}(V_{GS}-V_T)^2}_{I_D \text{ when } V_{DS}=0} \approx \lambda I_D$$

$$\lambda \approx \frac{1}{I_D}\frac{\Delta I_D}{\Delta V_{DS}}$$

Lambda is thus calculated with the drain current value at the y-axis extrapolated from the saturation region.

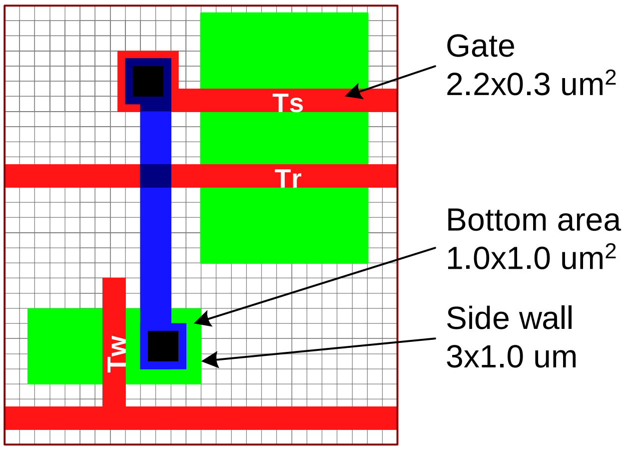


c. The threshold voltage can be estimated by plotting I_D against V_{GS} in the linear region or sqrt(I_D) against V_{GS} in the saturated region.



d. $I_{Dn}=k_n'/2*W_n/L_n*(3-V_{Tn})^2=1600e-6$ => $k_n' = 256e-6$
More accurate would be to use the value 1200e-6 which is the extrapolated value vhen $V_{DS}=0$, i.e. ID have no dependence on lambda.

8. The gate area, drain area and the drain side wall is shown in the figure



The total capacitance is thus

$$C_{tot} = 2.2\times0.3\times5.0 + 1.0\times1.0\times1.5 + 3\times1.0\times0.25 \text{ fF} = 3.3 + 1.5 + 0.75 \text{ fF} = 5.55 \text{ fF}$$

Comments: Only three sides of the sidewall are used since there is no diode towards the channel (page 111). The gate capacitance should be reduced to 2/3:s if saturation is assumed (table 3-4, page 109).